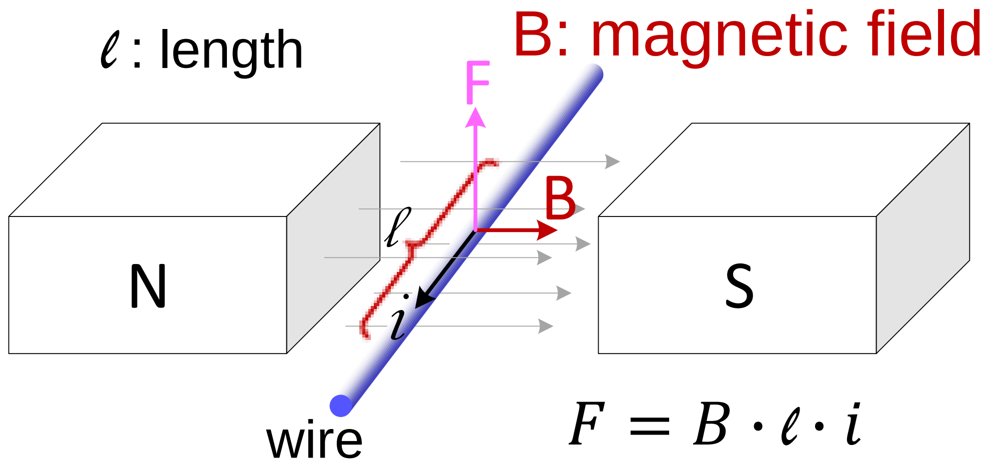
# Chapter 6: Electromechanical Systems

System Modeling and Simulation

(KON 301E)

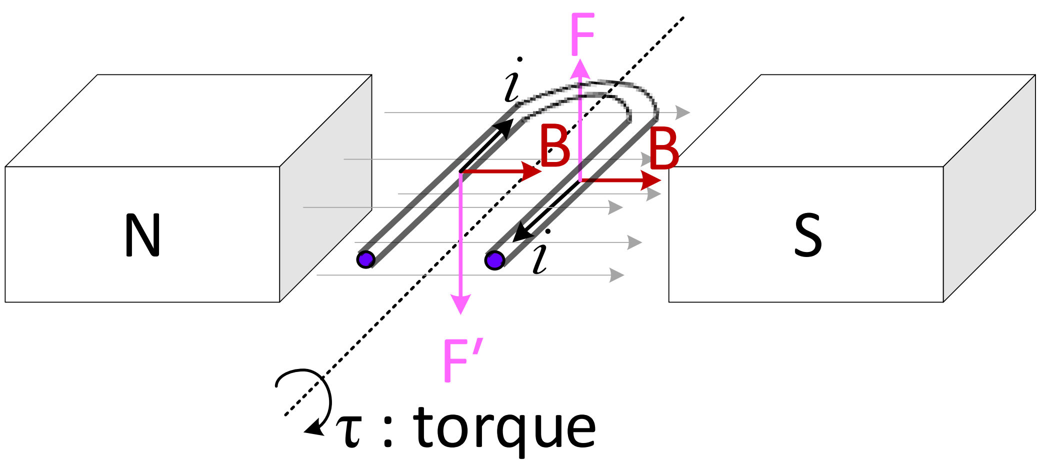
# Electromechanical Systems

- Behaviour of electric current in the presence of magnetic field

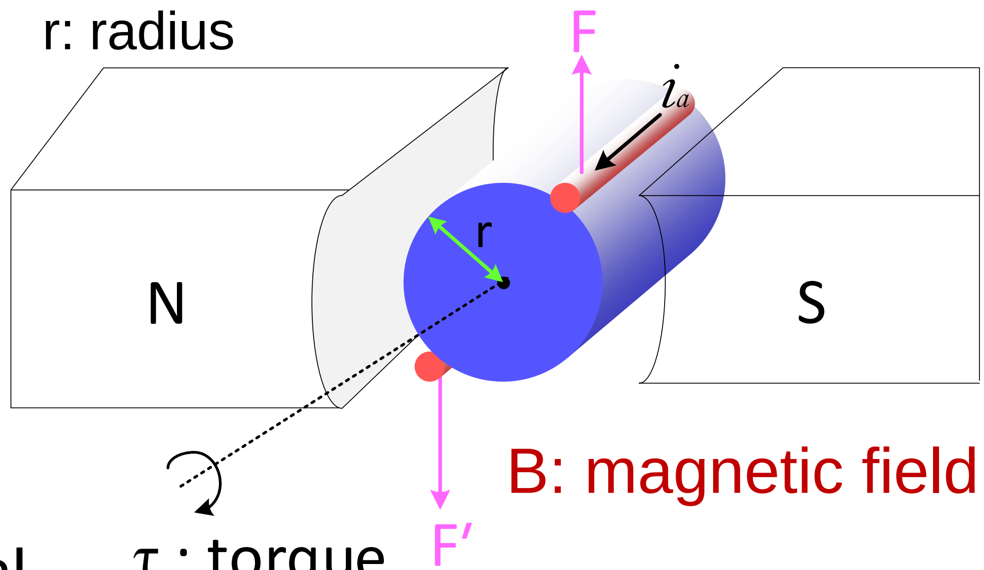



$$F = B \cdot \ell \cdot i$$

- For U shape conductor



# Electromechanical Systems




- Attach the wire to the rotor (≙rotating part)

rotor + wrapped wire ≡Armature

$i_a$ ≡armature current

- F: total force acts tangential

$$F = 2 \cdot B \cdot \ell \cdot i_a$$

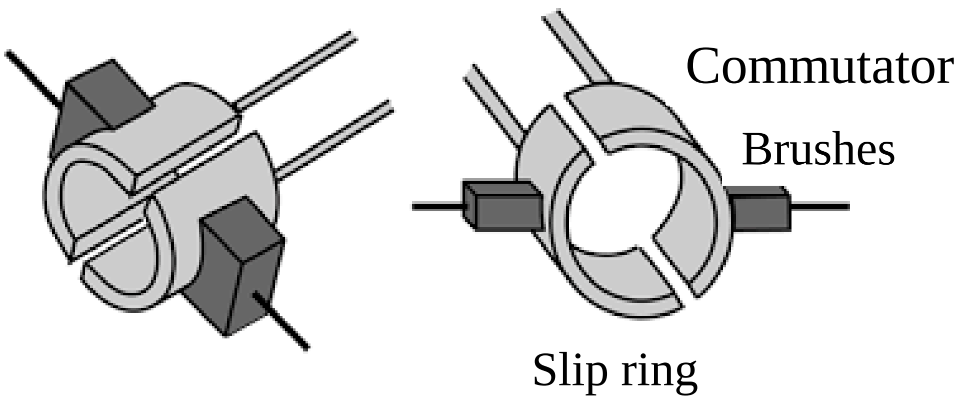
- Total torque induced on the rotor

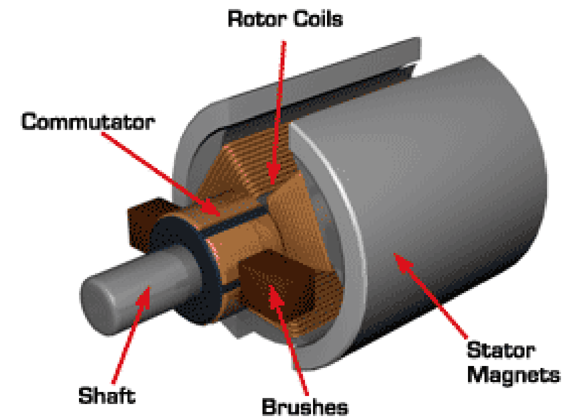
$$\tau_m = F \cdot r = \underbrace{r \cdot 2 \cdot \ell \cdot\ B}_{K_t} \cdot i_a \Rightarrow \boxed{\tau_m = K_t \cdot i_a}$$

$K_t$ : motor torque constant

# Electromechanical Systems

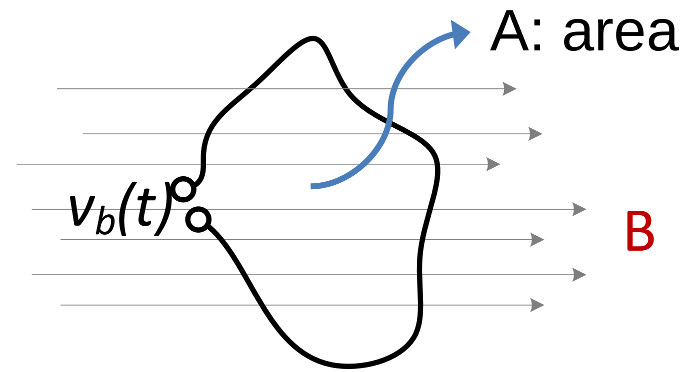
- PS: The current ($i_a$) is transferred to the rotating WINDINGs by SLIP RING + BRUSHES ≡ COMMUTATOR




- PS: Faraday Law
  - If a loop of wire is placed in a charging magnetic field, then a voltage is induced across the loop

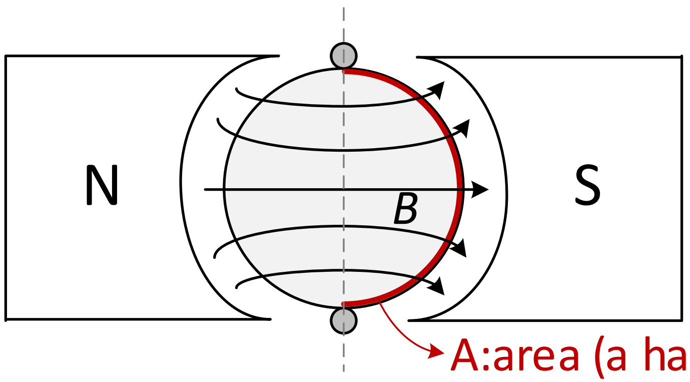
$$v_b(t) = -\frac{d\emptyset(t)}{dt} = -B\frac{dA(t)}{dt}$$



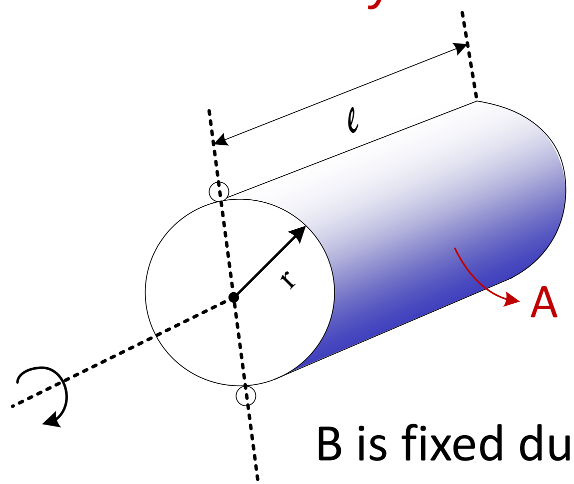

Φ= B.A : magnetic flux

# Electromechanical Systems

- Single loop winding around the armature
  - Example of voltage induction based on Faraday Law



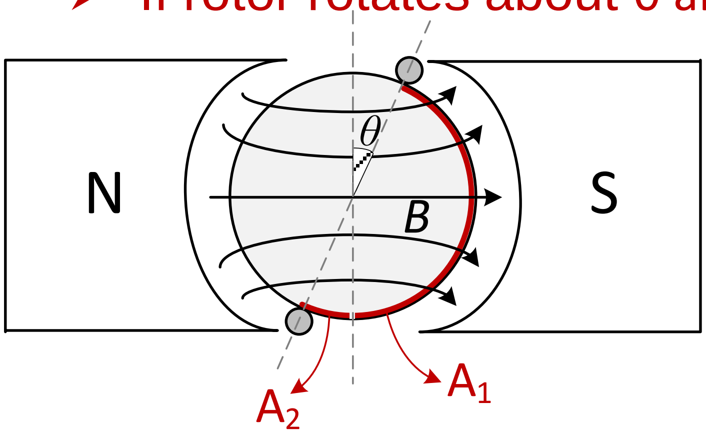

$\Phi = B.A$




$$A = \pi \cdot r \cdot \ell$$
*half a side*

B is fixed due to permanent magnets but A is <u>variable</u>!

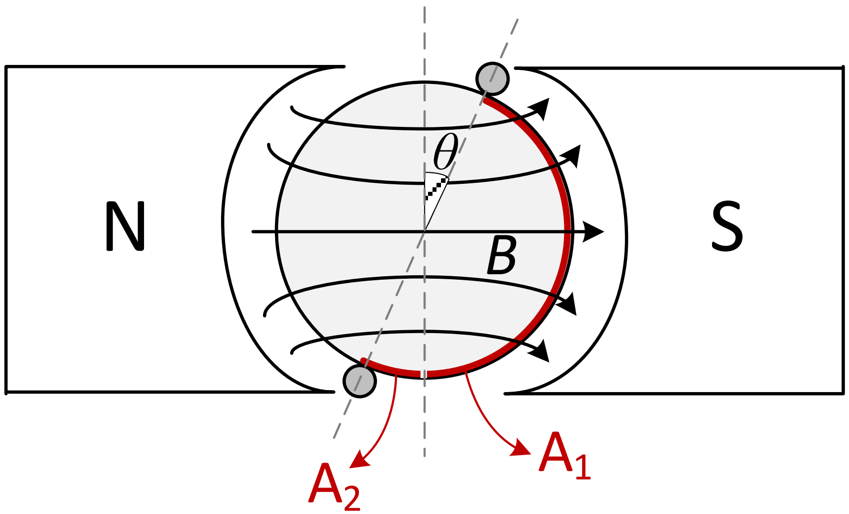
  - If rotor rotates about θ angle:




Magnetic field pass through $A_1$-$A_2$ area

$$\Phi = B \cdot (A_1 - A_2) \Rightarrow v_b(\mathrm{t}) = -B \cdot \frac{d(A_1 - A_2)}{dt}$$

# Electromechanical Systems




$$A_1 = r\cdot(\pi-\theta)\cdot\ell$$

$$A_2 = r\cdot\theta\cdot\ell$$

$$v_b(\mathrm{t}) = -B\cdot\frac{d(A_1-A_2)}{dt} = -B\cdot r\cdot\ell\cdot\frac{d(\pi-2\theta)}{dt} = 2\cdot B\cdot r\cdot\ell\cdot\underbrace{\frac{d\theta}{dt}}_{=\omega}$$

$$\boxed{v_b(\mathrm{t}) = 2\cdot B\cdot r\cdot\ell\cdot\omega = K_b\cdot\omega}$$ Back EMF voltage

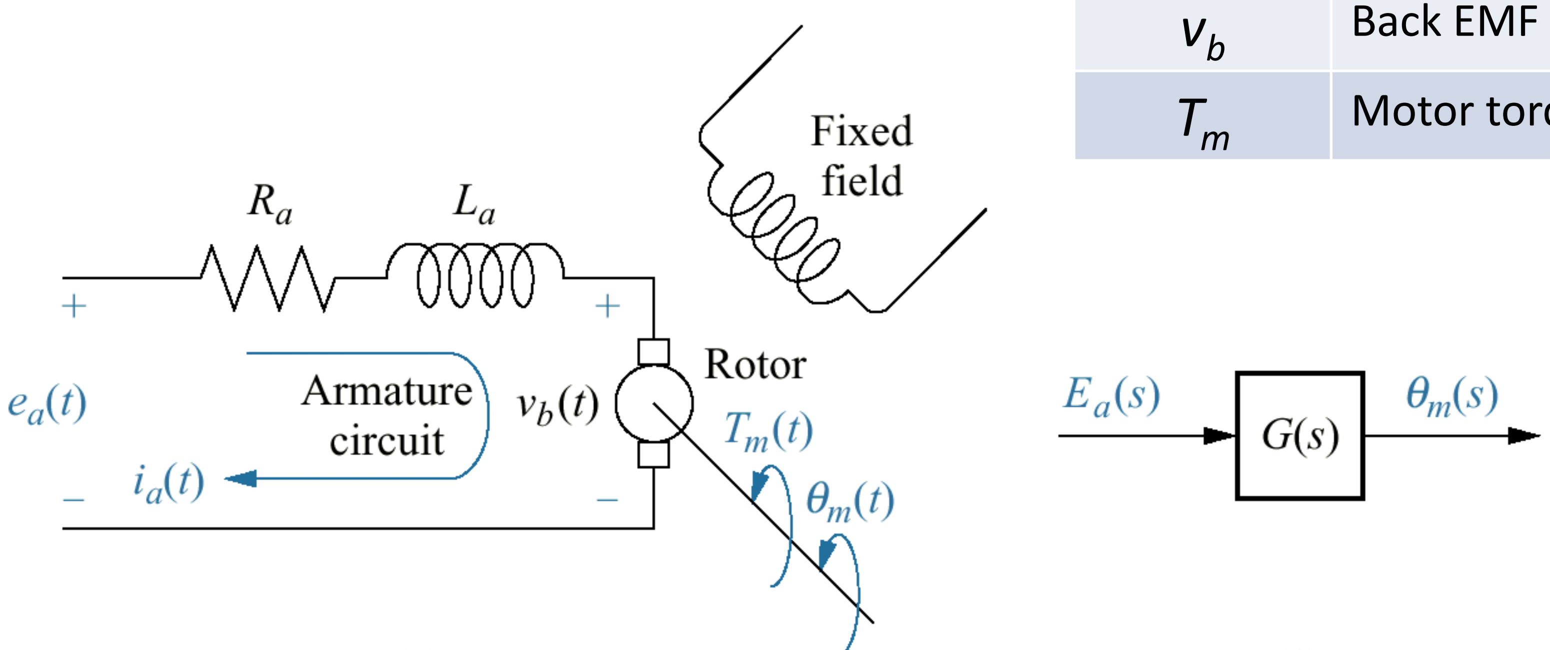
$K_b$: Back EMF constant

# Electromechanical Systems

## Modeling DC Motor

a. Schematic
b. Block diagram

| Parameter | Definition |
|---|---|
| $R_a$ | Armature resistance |
| $L_a$ | Armature inductance |
| $e_a$ | Armature voltage |
| $v_b$ | Back EMF voltage |
| $T_m$ | Motor torque |




(a) (b)

# Electromechanical Systems

- Electrical circuit:

$$e_a(t) - v_b(t) = R_a i_a(t) + L_a \frac{di_a(t)}{dt}$$

- Back Electromotor Force voltage

$$v_b(t) = K_b \omega_m(t) = K_b \frac{d\theta(t)}{dt}$$

- Torque-Current relation:

$$T_m(t) = K_t i_a(t)$$

# Electromechanical Systems

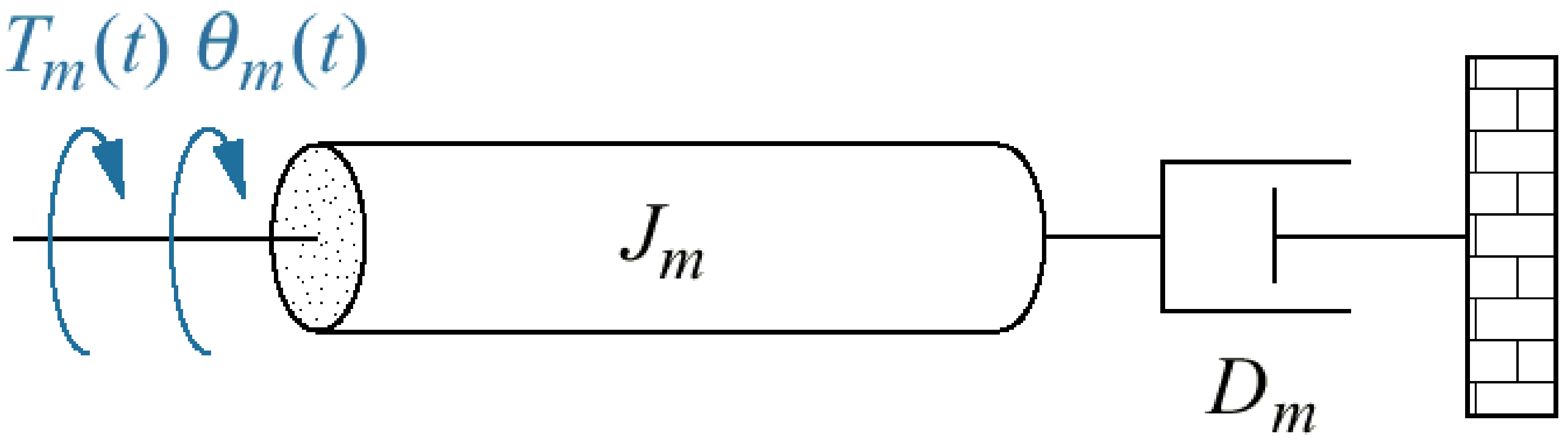
- In frequency domain:

$$E_a(s) = R_a I_a(s) + L_a s I_a(s) + V_b(s)$$

$$V_b(s) = K_b s\theta_m(s) \quad ; \quad T_m(s) = K_t I_a(s)$$

# Electromechanical Systems

Typical equivalent mechanical loading on a motor




PS: Sometimes $B_m$ is used instead of $D_m$

# Electromechanical Systems

- Mechanical Side:

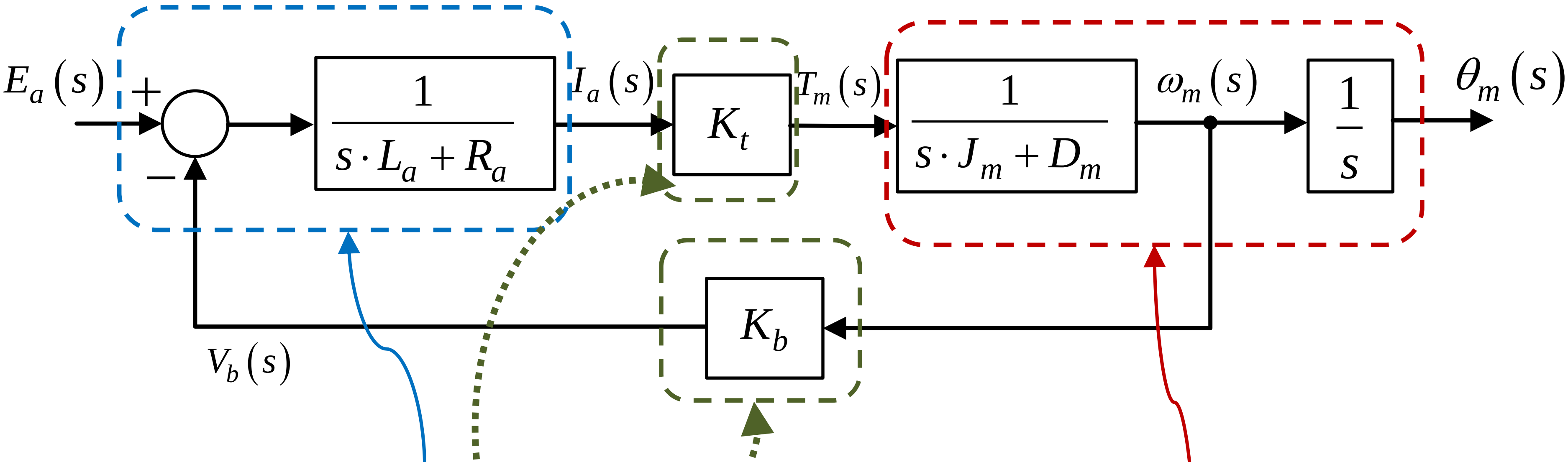
$$T_m(t) = J_m \frac{d^2\theta_m(t)}{dt^2} + D_m \frac{d\theta_m(t)}{dt}$$

- in frequency domain:

$$T_m(s) = J_m s^2 \theta_m(s) + D_m s \theta_m(s)$$

# Electromechanical Systems

- In block diagram (input: $E_a(s)$; output: $\theta_m(s)$)




$$E_a(s) = R_a I_a(s) + L_a s I_a(s) + V_b(s)$$

$$T_m(s) = J_m s^2 \theta_m(s) + D_m s \theta_m(s)$$

$$V_b(s) = K_b s \theta_m(s) \quad ; \quad T_m(s) = K_t I_a(s)$$

# Electromechanical Systems

- Transfer Function for angular position control:

$$\frac{\theta_m(s)}{E_a(s)} = \frac{K_t}{(sL_a + R_a)(J_m s^2 + D_m s) + K_b K_t s}$$

- If $L_a \sim 0$ :

$$\frac{\theta_m(s)}{E_a(s)} = \frac{K_t/(R_a J_m)}{s\left[s + \frac{1}{J_m}\left(D_m + \frac{K_b K_t}{R_a}\right)\right]}$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{K}{s(s+\alpha)}$$

# Electromechanical Systems

- Transfer Function for angular speed control:

$$\frac{\omega_m(s)}{E_a(s)} = \frac{K_t}{(sL_a + R_a)(J_m s + D_m) + K_b K_t}$$

- If $L_a \sim 0$ :

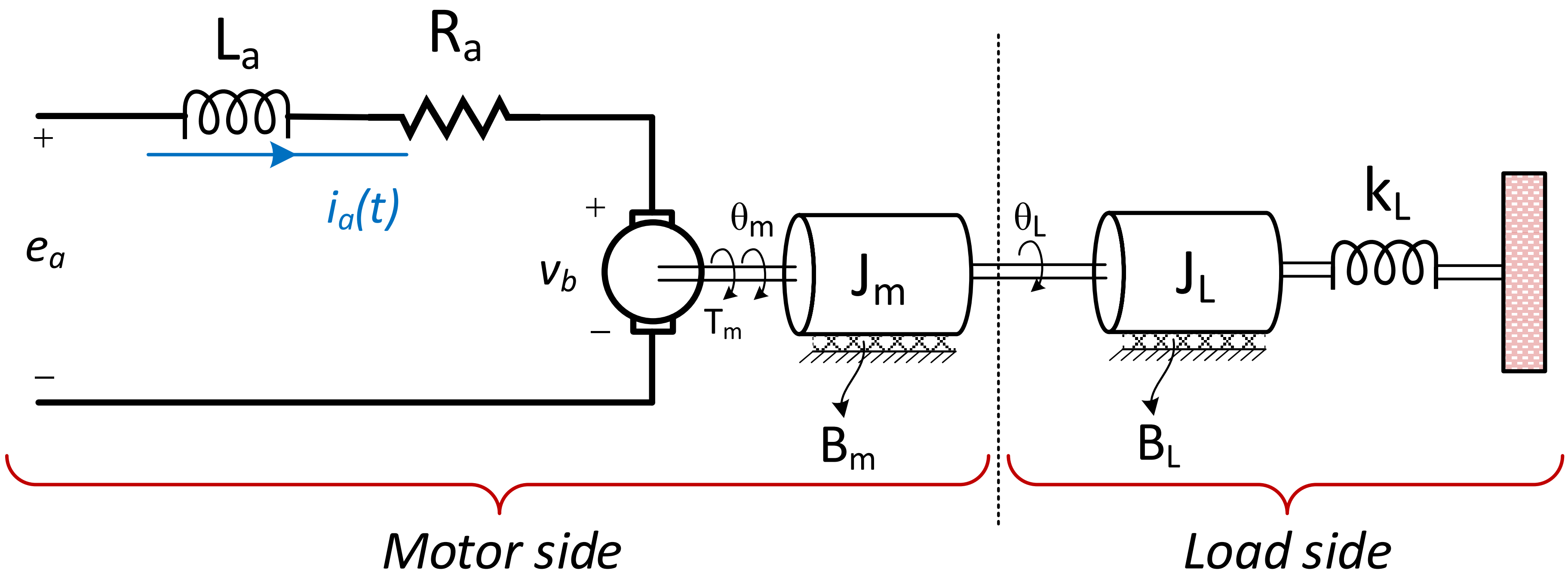
$$\frac{\omega_m(s)}{E_a(s)} = \frac{K_t/(R_a J_m)}{s + \frac{1}{J_m}\left(D_m + \frac{K_b K_t}{R_a}\right)}$$

$$\frac{\omega_m(s)}{E_a(s)} = \frac{K}{s + \alpha}$$

# Electromechanical Systems

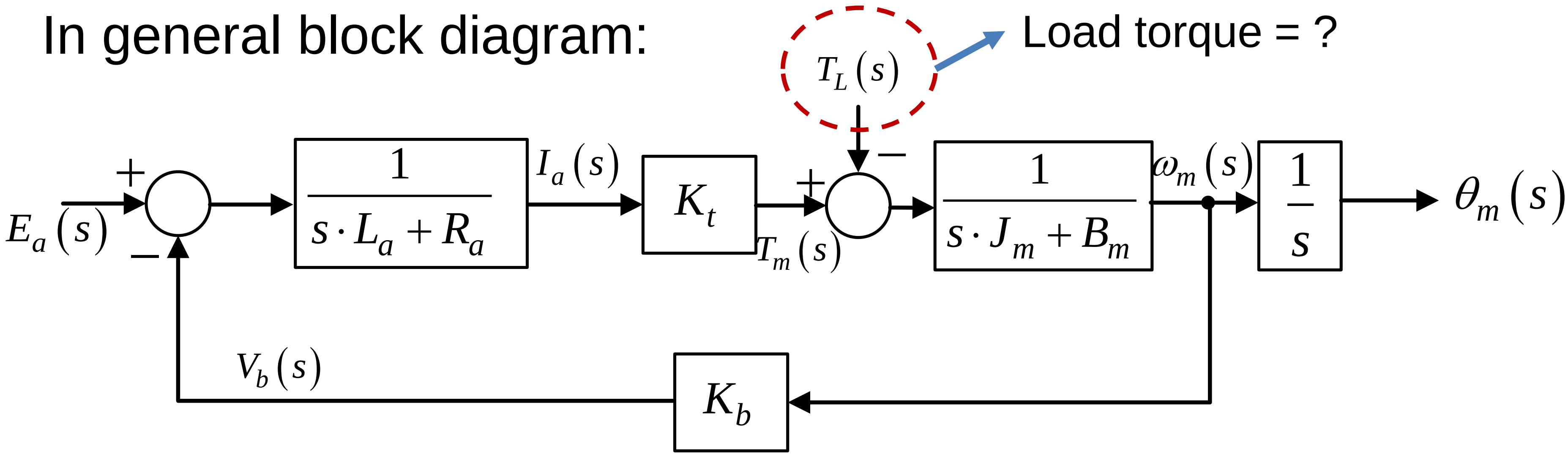
Modeling a DC motor driving a rotational mechanical load

- with rigid coupling: $\theta_m = \theta_L$



# Electromechanical Systems

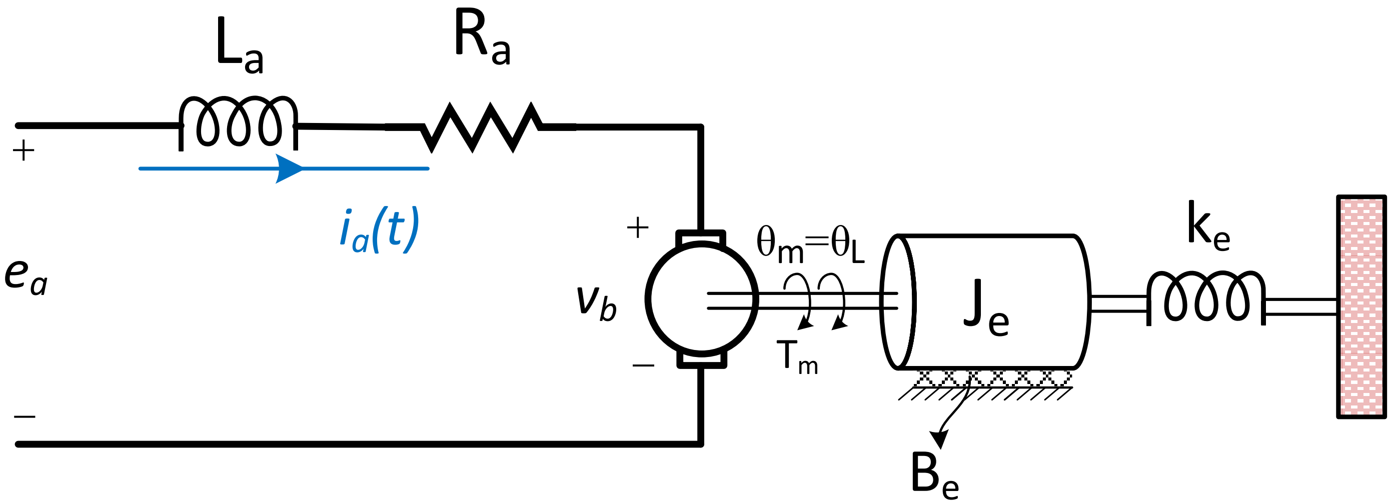
In general block diagram:




PS: It is possible to find '$T_L$' and design a block diagram but it is not preferred!

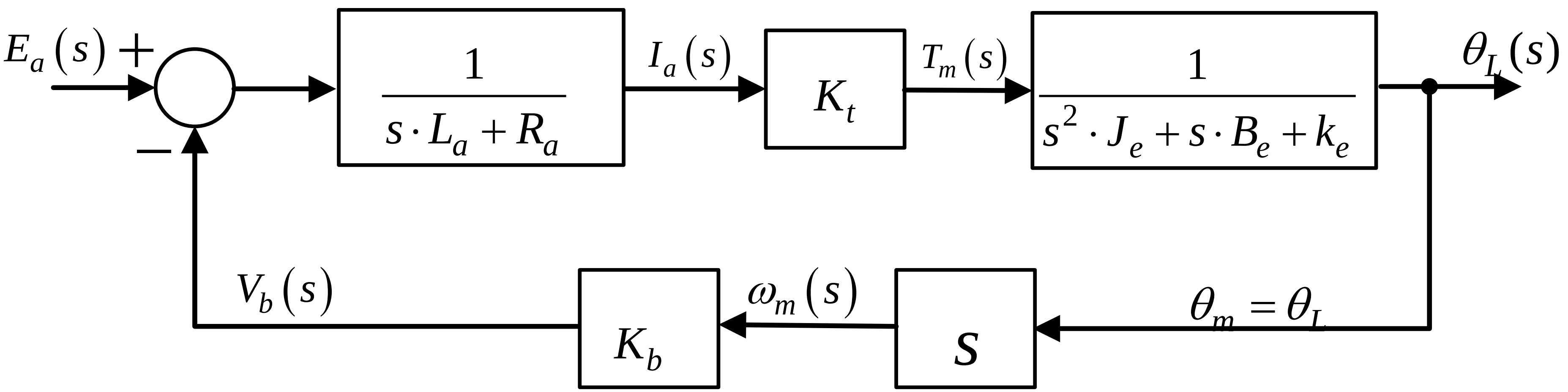
- Write 'equivalent mechanical loading on a motor' in terms of
  - equivalent inertia at armature → $J_e$
  - equivalent friction at armature → $B_e$
  - equivalent spring at armature → $k_e$

# Electromechanical Systems




$$J_e = J_m + J_L$$
$$B_e = B_m + B_L$$
$$k_e = k_L$$

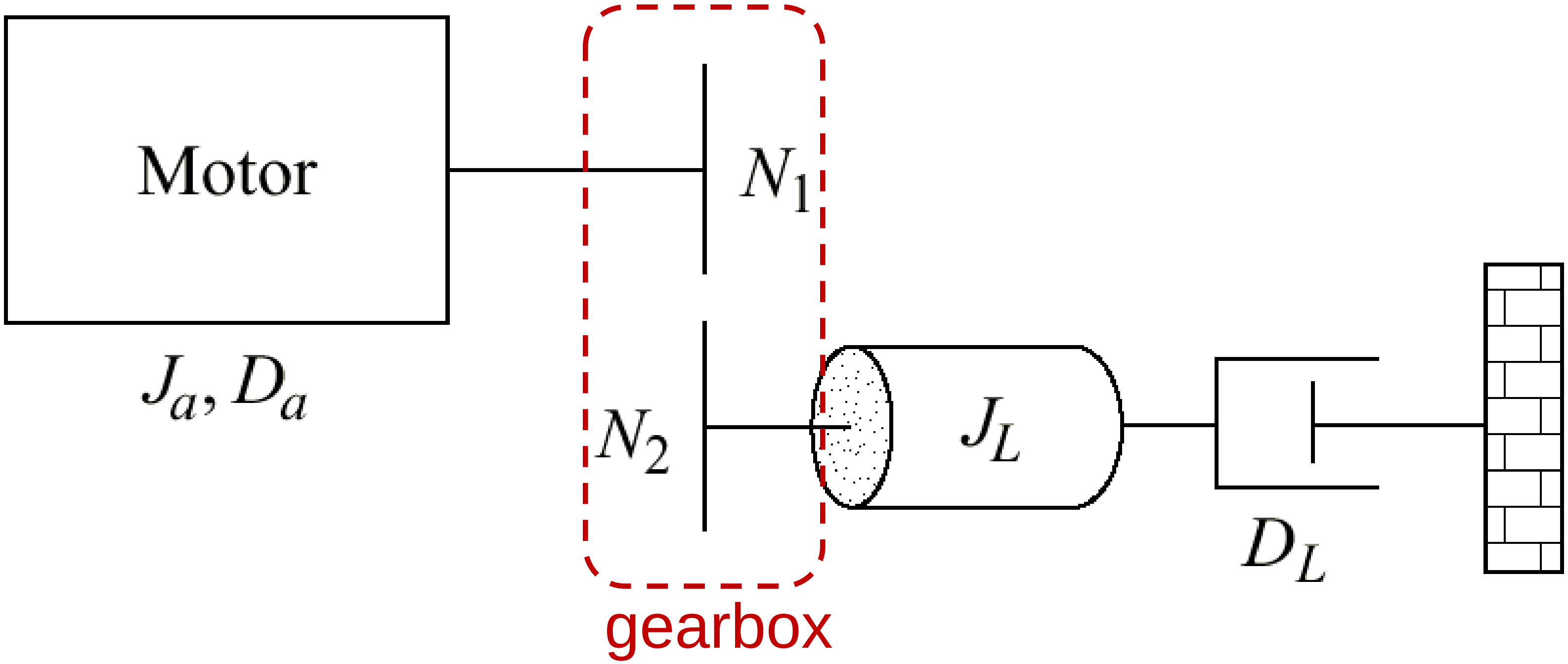
$$T_m(s) = \left(s^2 \cdot J_e + s \cdot B_e + k_e\right) \cdot \theta_L(s)$$



# Electromechanical Systems

Modeling a DC motor driving a rotational mechanical load

- with a gearbox



# Electromechanical Systems

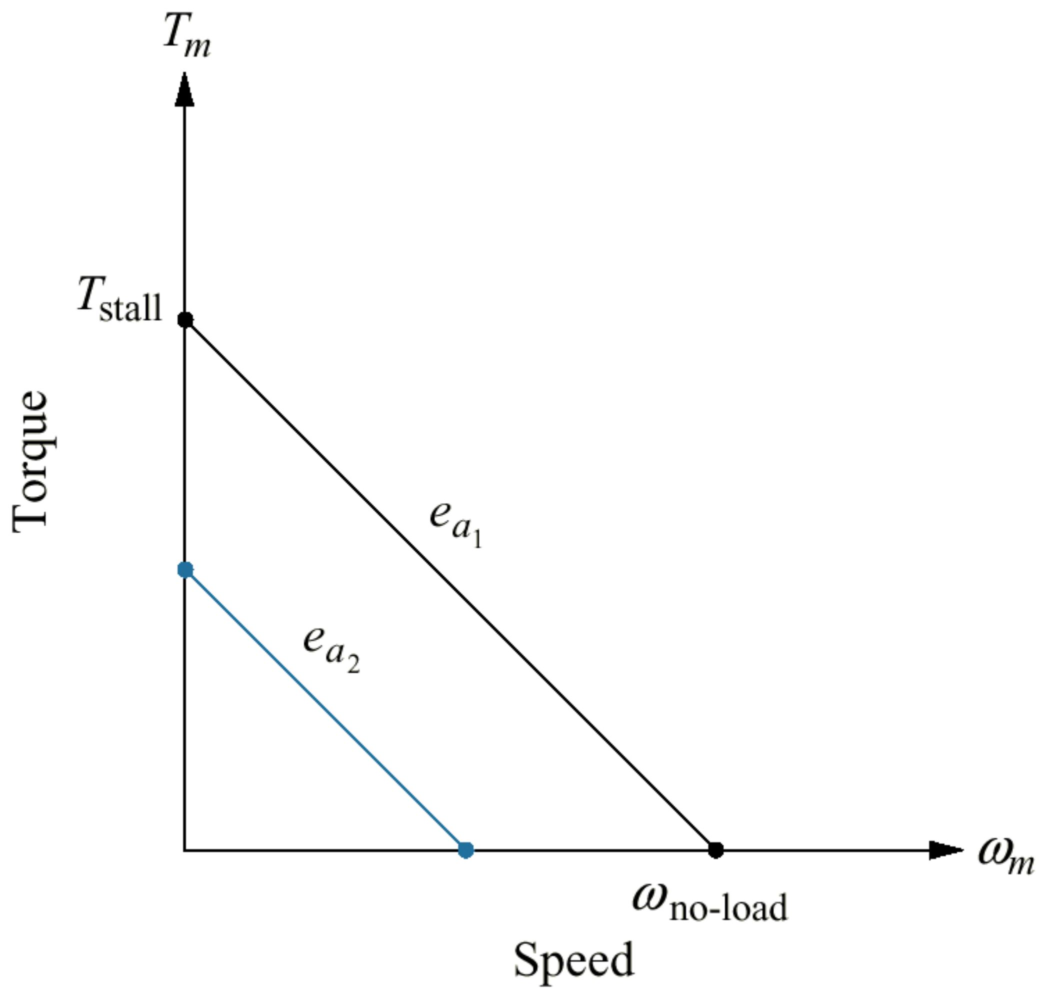
- Equivalent inertia and damping at the armature:

$$J_m = J_a + J_L \left( \frac{N_1}{N_2} \right)^2 \quad ; \quad D_m = D_a + D_L \left( \frac{N_1}{N_2} \right)^2$$

- Electrical parameters are defined by **using dynamometers**

# Electromechanical Systems

Torque-speed curves with an armature voltage, $e_a$ , as a parameter



# Electromechanical Systems

- Electrical equation:

$$E_a(s) = R_a I_a(s) + \underbrace{L_a\, s I_a(s)}_{\approx 0} + V_b(s)$$

$$I_a(s) = \frac{1}{K_t} T_m(s) \qquad\qquad V_b(s) = K_b s \theta_m(s)$$

$$\frac{R_a}{K_t} T_m(s) + K_b \underbrace{s\theta_m(s)}_{\omega_m(s)} = E_a(s)$$

# Electromechanical Systems

- Electrical equation:

$$T_m(s) = -\frac{K_b K_t}{R_a}\omega_m(s) + \frac{K_t}{R_a}E_a(s)$$

$$\downarrow \mathcal{L}^{-1}$$

$$\underbrace{T_m(t) = -\frac{K_b K_t}{R_a}\omega_m(t) + \frac{K_t}{R_a}e_a(t)}_{\text{a straight line}}$$

# Electromechanical Systems

- For $\omega_m = 0$ :

$$T_m = T_{stall} = \frac{K_t}{R_a} e_a \qquad : \text{stall torque}$$

- For $T_m = 0$ :

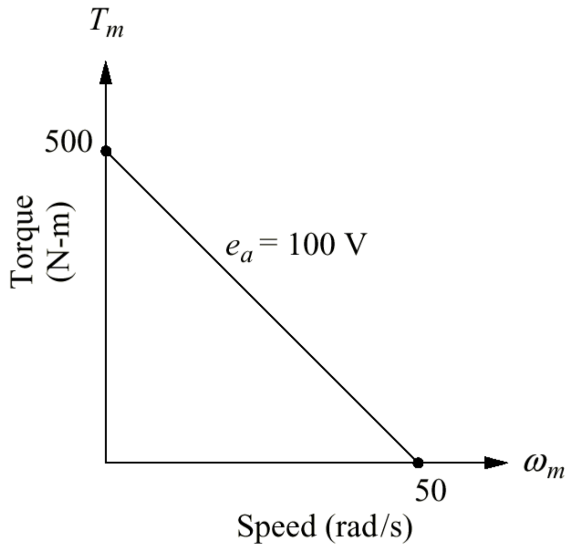
$$\omega_m = \omega_{no-load} = \frac{1}{K_b} e_a \qquad : \text{no-load speed}$$

- Electrical constants of the motor :

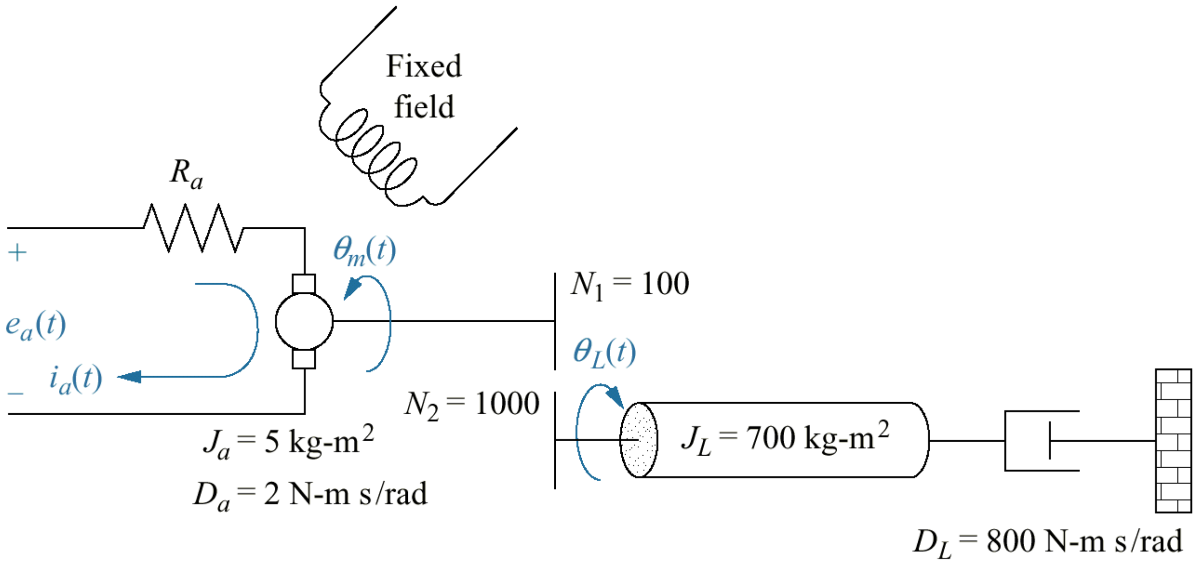
$$\frac{K_t}{R_a} = \frac{T_{stall}}{e_a} \quad ; \qquad K_b = \frac{e_a}{\omega_{no-load}}$$

# Electromechanical Systems

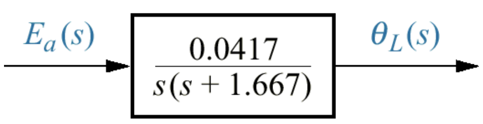
**a.** DC motor and load;
**b.** torque-speed curve;
**c.** block diagram;




(*b*)




(*a*)




(*c*)

- Solution:

$$J_m = J_a + J_L\left(\frac{N1}{N2}\right)^2 = 5 + 700\left(\frac{1}{10}\right)^2 = 12$$

$$D_m = D_a + D_L\left(\frac{N1}{N2}\right)^2 = 2 + 800\left(\frac{1}{10}\right)^2 = 10$$

$$e_a = 100 \qquad \mathrm{T}_{\mathrm{stall}} = 500 \qquad \omega_{\text{no-load}} = 50$$

- 

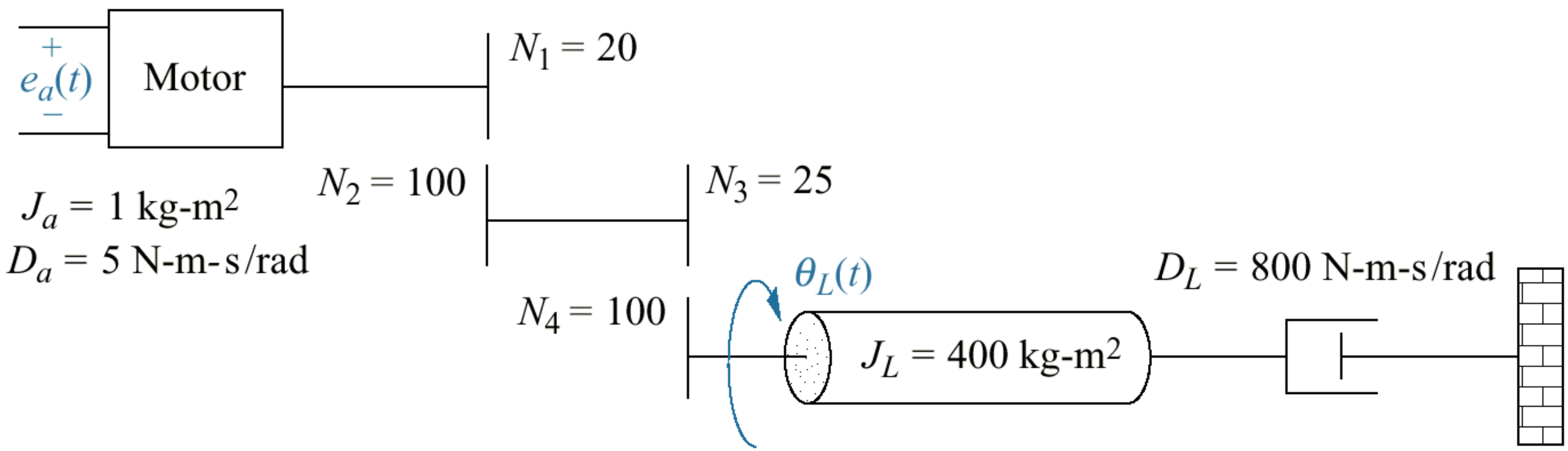
$$\frac{K_t}{R_a} = \frac{T_{stall}}{e_a} = \frac{500}{100} = 5 \quad ; \quad K_b = \frac{e_a}{\omega_{no-load}} = \frac{100}{50} = 2$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{5/12}{s\left(s + \frac{1}{12}(10 + 5\cdot 2)\right)} = \frac{0.417}{s(s+1.667)}$$

$$\theta_m(s) = \frac{N_2}{N_1}\theta_L(s) = 10\,\theta_L(s) \quad \Rightarrow \quad \frac{\theta_L(s)}{E_a(s)} = \frac{0.0417}{s(s+1.667)}$$

# Electromechanical Systems

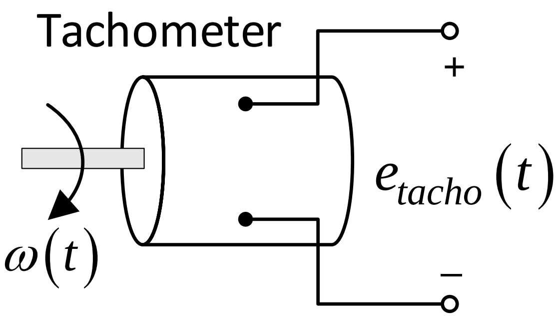
**Example:** Skill-Assessment Exercise for DC Motor



# Electromechanical Systems

**Tachometers:** A voltage generator, gives a voltage output proportional to the input angular velocity.

- an angular rate sensor!
- a DC machine




$$e_{tacho}(t) = K_{tacho} \cdot \omega(t)$$

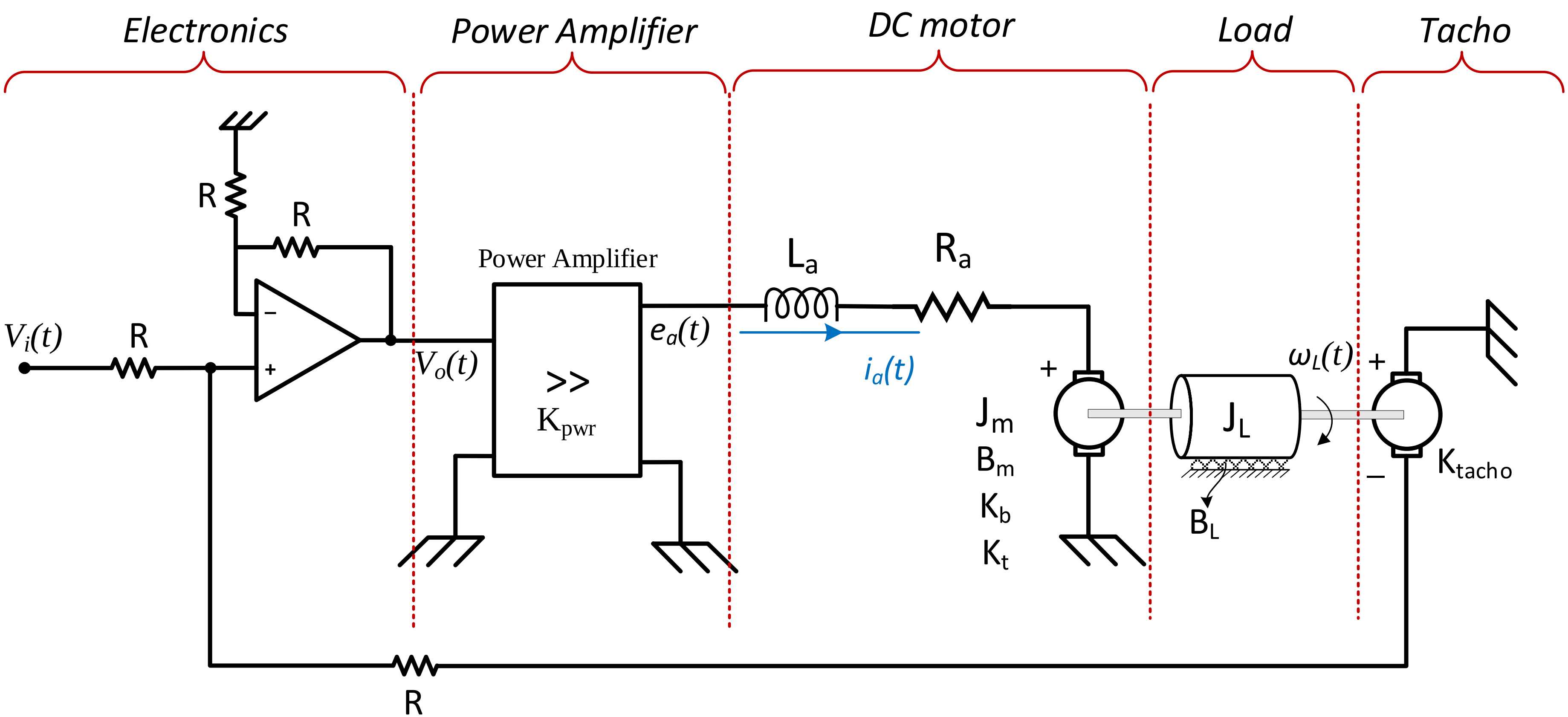
$$\Rightarrow E_{tacho}(s) = K_{tacho} \cdot \omega(s)$$

$\boxed{K_{tacho} : \text{tachometer constant}}$

$$\omega(s) \longrightarrow \boxed{K_{tacho}} \longrightarrow E_{tacho}(s)$$

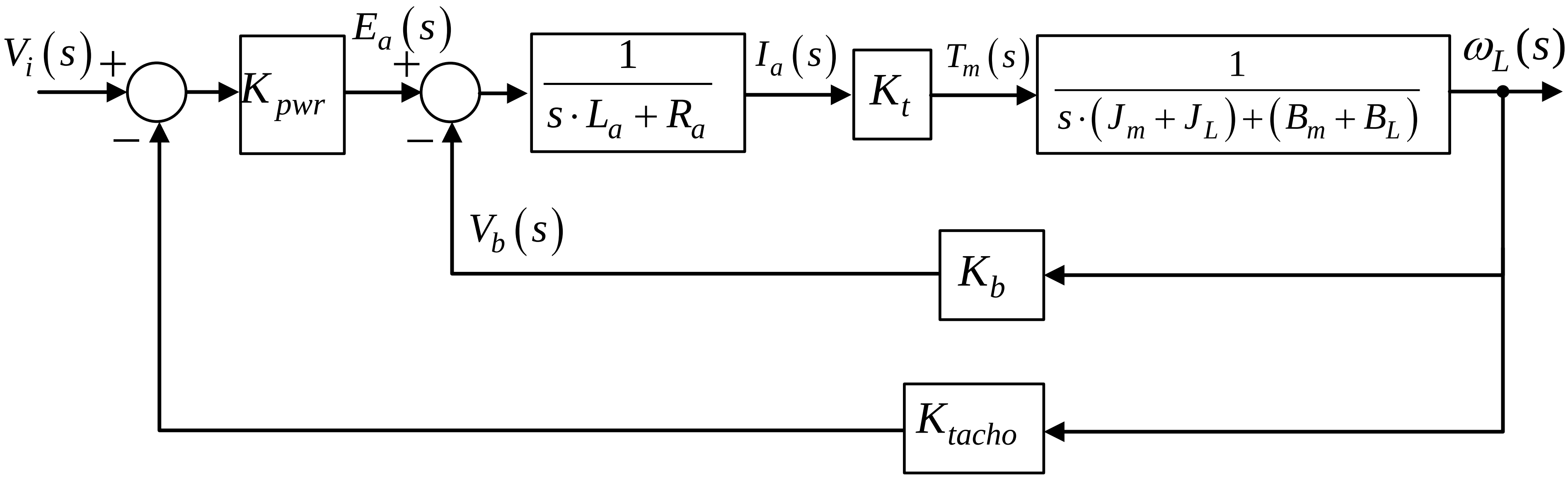
# Electromechanical Systems

**Example:** For the system below, draw block diagram and write $\omega_L(s)/V_i(s)$ transfer function



# Electromechanical Systems

Block diagram of the system:




Then, transfer function can be derived via block reduction

# Next class…

- Fluid Systems

have any questions or concerns!